

M. Hohle:

Physics 77: Introduction to Computational Techniques in Physics

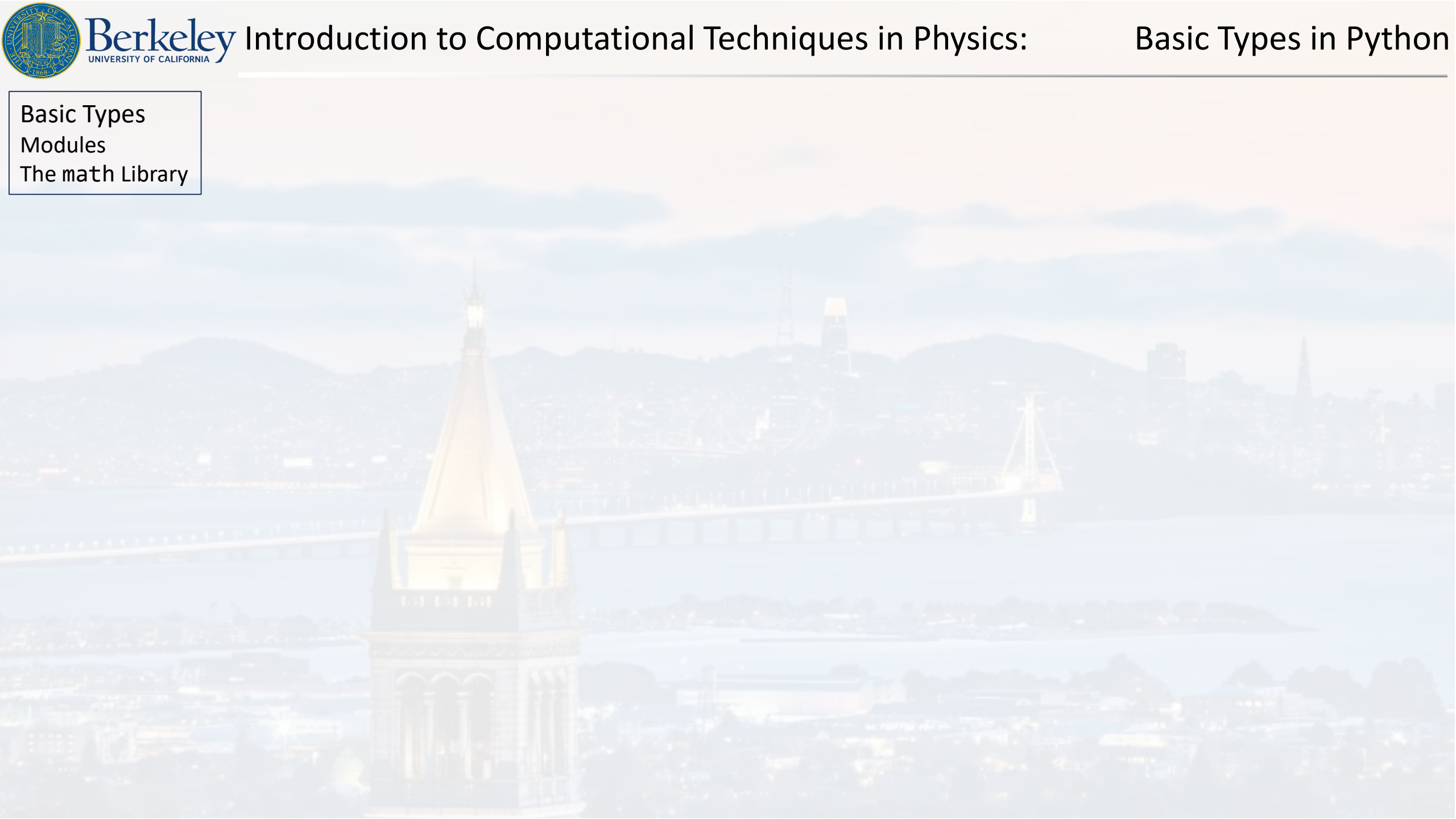




Week	Date	Topic
1	Aug 26 th	Programming Environment & UIs for Python. Programming Fundamentals
2	Sep 2nd	Basic Types in Python
3	Sep 9 th	Parsing, Data Processing and File I/O
4	Sep 16 th	Visualization: Matplotlib & Seaborn
5	Sep 23 rd	Functions, Map & Lambda
6	Sep 30 th	Random Numbers & Probability Distributions
7	Oct 7 th	Interpreting Measurements & Introduction to Bayesian Statistics
8	Oct 14 th	Numerical Integration and Differentiation
9	Oct 21 st	Root finding, Interpolation
10	Oct 28 th	Systems of Linear Equations
11	Nov 4 th	Introduction to Ordinary Differential Equations (ODEs)
	Nov 11 th	<i>Veteran's Day</i>
12	Nov 18 th	Stability of ODEs, Runge-Kutta, Examples
	Nov 25 th	<i>Thanksgiving</i>
13	Dec 2 nd	Final Project Presentations



Basic Types
Modules
The math Library





Basic Types

Modules

The math Library

The zoo of types

numeric: `int`, `float`, `complex` `5`, `5.55`, `(5+5j)`

strings: `str` `'this is a string'`, `"this is a string"`

iteratable

sequence: `list`, `tuple`, `range` `my_tuple = (3, 'a', [2,3,4,5])`
`range(10)`

mutable

`my_list = [1, 2, 'a']`

mapping: `dict` `my_dict = {1: 'a', 2: 'b'}`

mapping: `set` `my_set = {1, 2, 'a'}`

boolean: `True` `False`

none type: `None`

callable: functions, methods, classes `def`, `class`, `map`, `lambda`

modules: `from my_module import my_method as my_alias`



Basic Types

strings

Modules

The math Library

(data) types: data values, with specific **set of possible values** and set of **allowed operations**

```
my_string = 'this is a string'
```

A variable called `my_string`, that contains the *string* `'this is a string'`

```
type(my_string)  
str
```

The **type** of `my_string` is `str` (= “string”)

```
my_number = 5
```

A variable called `my_number`, that contains an *integer* `5`

```
type(my_number)  
int
```

The **type** of `my_number` is `int` (= “integer”)



Basic Types

strings

Modules

The math Library

(data) types: data values, with specific **set of possible values** and set of **allowed operations**

```
my_string = 'this is a string'
```

```
my_string = '5'
```

```
type(my_string)  
str
```

```
my_number = 5
```

```
my_number = Ab
```

```
In [49]: my_number = Ab  
Traceback (most recent call last):
```

```
Cell In[49], line 1  
    my_number = Ab
```

```
NameError: name 'Ab' is not defined
```

```
my_string = 5
```

```
type(my_string)  
int
```

```
my_number = 'Ab'
```

```
type(my_number)  
str
```




Basic Types

strings

Modules

The math Library

(data) types: data values, with specific **set of possible values** and set of **allowed operations**

```
my_string = 'this is a string'
```

```
2*my_string
```

```
my_string + my_string
```

```
'this is a stringthis is a string'
```

```
my_number = 5
```

```
my_string/3
```

Traceback (most recent call last):

Cell In[56], line 1

```
my_string/3
```

TypeError unsupported operand type(s) for /: 'str' and 'int'

type error: operation is invalid for this specific type!



Basic Types

strings

Modules

The math Library

(data) types: data values, with specific **set of possible values** and set of **allowed operations**

```
my_string = 'this is a string'
```

```
2*my_string
```

```
my_string + my_string
```

```
'this is a stringthis is a string'
```

```
my_number = 5
```

```
2*my_number
```

```
my_number + my_number
```

```
10
```

```
my_string/3
```

Traceback (most recent call last):

```
Cell In[56], line 1  
    my_string/3
```

TypeError: unsupported operand type(s) for /: 'str' and 'int'



Basic Types

strings

Modules

The math Library

(data) types: data values, with specific **set of possible values** and set of **allowed operations**

```
my_string = 'this is a string'
```

```
2*my_string
```

```
my_string + my_string
```

```
'this is a stringthis is a string'
```

```
my_number = 5
```

```
2*my_number
```

```
my_number + my_number
```

```
10
```

The fact that we can use the same operator (here +) for different types is called **operator overload**



Basic Types

strings

Modules

The math Library

(data) types: data values, with specific **set of possible values** and set of **allowed operations**

```
my_string = 'this is a string'
```

```
my_string.  
capitalize  
casefold  
center  
count  
encode  
endswith  
expandtabs  
find  
format  
format_map  
index
```

```
my_number = 5
```

```
my_number.  
as_integer_ratio  
bit_count  
bit_length  
conjugate  
denominator  
from_bytes  
imag  
numerator  
real  
to_bytes
```



Basic Types

strings

Modules

The math Library

when to use:

labels and titles of plots
paths and file names
error messages

```
string1 = 'Hello Students'  
string2 = ', how are you'
```

```
string12 = string1 + string2  
'Hello Students, how are you'
```

concatenating is incredibly easy!

```
S = 'abc'
```

```
3*S  
'abcabcabc'
```

```
string12[2:6]
```

[1, 5, 0, -3]

slices:

0 1 2 3 4

slicing



Basic Types

strings

Modules

The math Library

```
string12[2:6]
```

[1, 5, 0, -3]

slices: 0 1 2 3 4

[1, 5, 0, -3]

index: 0 1 2 3

index: -4 -3 -2 -1

```
string12[-1]
```

```
string12[1:]
```

```
string12[:-1]
```

indexing

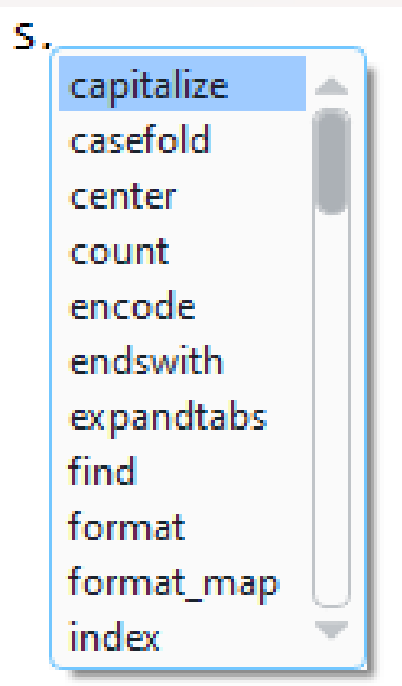


Basic Types

strings

Modules

The math Library



try some of the functions like

```
S.count()
```

```
S.find()
```



Basic Types

int

Modules

The math Library

```
my_int = 5
```

```
type(my_int)  
Int
```

```
my_float = 5.0
```

```
type(my_float)  
float
```

```
type(10/5)  
float
```

check out:

```
5**2
```

```
36**0.5
```

```
5//3
```

```
6//3
```

```
5%3
```

```
6%3
```

```
type(str(6))
```

```
my_int.  
as_integer_ratio  
bit_count  
bit_length  
conjugate  
denominator  
from_bytes  
imag  
numerator  
real  
to_bytes
```




Basic Types

int

Modules

The math Library

```
c1 = (-1)**0.5
```

```
print(c1)
```

```
(6.123233995736766e-17+1j)
```

machine epsilon: 2.2e-16

complex unit i , in Python: j

```
c2 = complex(real = 5, imag = -4)
```

```
print(c2)
```

```
(5-4j)
```

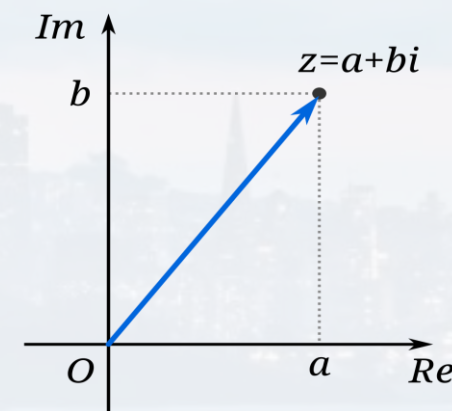
check out:

```
c2**2
```

```
c2**0.25
```

```
c2.imag
```

```
c2.real
```





Basic Types

boolean

Modules

The math Library

when to use:

checking if statement is true or false

checking if variable has a value/ is of a type

catching up error messages

```
my_number = 5
```

assigning a value to a variable

```
my_number == 5
```

True

logical operations
return **boolean**
statement

```
my_number > 5
```

False

```
my_number != 5
```

False

value of a variable equal to another value

```
type(my_number > 5)
```

bool



Basic Types

boolean

Modules

The math Library

when to use:

checking if statement is true or false

checking if variable has a value/ is of a type

catching up error messages

```
my_number = 5  
string    = 'Python is great!'
```

```
isinstance(my_number, int)  
True
```

```
isinstance(my_number, float)  
False
```

```
isinstance(string, str)  
True
```

```
isinstance(True, bool)  
True
```

checking if variable has a value
(What do you expect the output is?)

```
bool(None)
```

```
bool(0)
```

```
bool("")
```

```
bool(())
```

```
bool([])
```

```
bool({})
```

```
bool(False)
```

```
bool(True)
```




Basic Types

boolean

Modules

The math Library

when to use:

- checking if statement is true or false
- checking if variable has a value/ is of a type
- catching up error messages

Let's run the following code together and try to understand, what it does:

```
def CheckTimes2(var = None):  
    if not bool(var):  
        print('You need an input!')  
    elif isinstance(var, float):  
        return 2*var  
    elif isinstance(var, int):  
        return 2*var  
    elif isinstance(var, str):  
        return 2*var  
    else:  
        print('not possible to multiply ' + str(type(var)))
```



The zoo of types



numeric: `int`, `float`, `complex`

`5`, `5.55`, `(5+5j)`

strings: `str`

`'this is a string'`, `"this is a string"`

iteratable

sequence: `list`, `tuple`, `range`

`my_tuple = (3, 'a', [2,3,4,5])`
`range(10)`

mutable

`my_list = [1, 2, 'a']`

mapping: `dict`

`my_dict = {1: 'a', 2: 'b'}`

mapping: `set`

`my_set = {1, 2, 'a'}`



boolean:

`True` `False`

none type:

`None`

callable: functions, methods, classes

`def`, `class`, `map`, `lambda`

modules:

`from my_module import my_method as my_alias`



Basic Types

list

Modules

The math Library

when to use:

“default” type in Python

storing variables of different types in one object

error messages

```
L1 = [1, 2, 'a', complex(3,4)]
```

```
type(L1)  
list
```

```
print(2*L1)  
[1, 2, 'a', (3+4j), 1, 2, 'a', (3+4j)]
```

recall: operator overload
(see '*strings*')

```
print(L1 + L1)  
[1, 2, 'a', (3+4j), 1, 2, 'a', (3+4j)]
```

```
L1[2:4]
```

```
[ 1, 2, 'a', (3+4j) ]
```

slices:

0 1 2 3 4

slicing is identical too



Basic Types

list

Modules

The math Library

```
L1 = [1, 2, 'a', complex(3,4)]
```

```
L2 = L1
```

```
print(L1[2], L2[2])
```

```
a a
```

```
L1[2] = 'b'
```

```
print(L1[2], L2[2])
```

```
b b
```

```
L2[2] = 'a'
```

```
print(L1[2], L2[2])
```

```
a a
```

Even though we only changed L1, it affected L2 too!

Lists are **mutable**!



Basic Types

list

Modules

The math Library

```
L1 = [1, 2, 'a', complex(3,4)]
```

```
L2 = L1.copy()
```

```
print(L1[2], L2[2])
```

a a

```
L1[2] = 'b'
```

```
print(L1[2], L2[2])
```

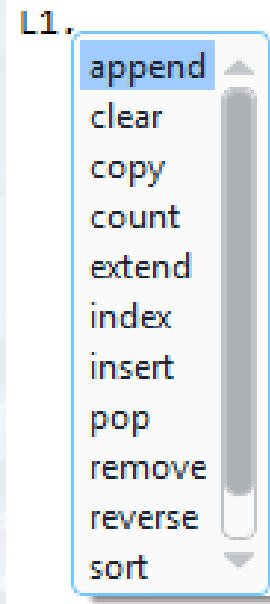
b a

```
L2[2] = 'c'
```

```
print(L1[2], L2[2])
```

b c

not passing mutability to L2





Basic Types

dict

Modules

The math Library

when to use:

keyword arguments (**kwargs) in functions
making code more compact (vs control structures)

```
D1 = dict(
```

```
    name = "John Doe",  
    DoB  = 'March 2nd, 2005',  
    grades = ["A+", 'B', 'A-'],  
    year  = 2024
```

key

value

creating a dictionary using
the **constructor** `dict`

```
D2 =
```

```
{    'name': "John Doe",  
    'DoB': 'March 2nd, 2005',  
    'grades': ["A+", 'B', 'A-'],  
    'year': 2024
```

```
}
```

creating a dictionary using
`{ }`



Basic Types

dict

Modules

The math Library

```
type(D2)
```

```
dict
```

```
print(D2)
```

```
{'name': 'John Doe', 'DoB': 'March 2nd, 2005', 'grades': ['A+', 'B', 'A-'],  
'year': 2024}
```

```
D2['name']
```

```
'John Doe'
```

```
D2.keys()
```

```
dict_keys(['name', 'DoB', 'grades', 'year'])
```

```
D2.values()
```

```
dict_values(['John Doe', 'March 2nd, 2005', ['A+', 'B', 'A-'], 2024])
```



Basic Types

dict

Modules

The math Library

```
print(D2)
{'name': 'John Doe', 'DoB': 'March 2nd, 2005', 'grades': ['A+', 'B', 'A-'],
'year': 2024}
```

```
D2.update(Address = '134 Street, Home')
```

adding keys

```
D2
{'name': 'John Doe',
'DoB': 'March 2nd, 2005',
'grades': ['A+', 'B', 'A-'],
'year': 2024,
'Address': '134 Street, Home'}
```



Basic Types

dict

Modules

The math Library

```
print(D2)
{'name': 'John Doe', 'DoB': 'March 2nd, 2005', 'grades': ['A+', 'B', 'A-'],
 'year': 2024}
```

```
D2['name'] = 'Tony Clifton'
```

updating values

```
D2
{'name': 'Tony Clifton',
 'DoB': 'March 2nd, 2005',
 'grades': ['A+', 'B', 'A-'],
 'year': 2024,
 'Address': '134 Street, Home'}
```




Basic Types

dict

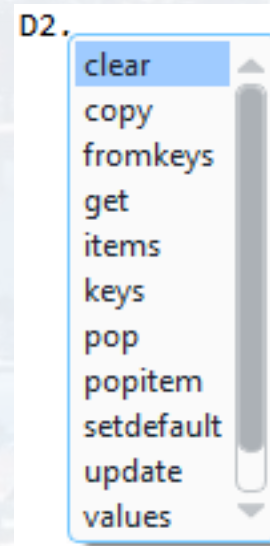
Modules

The math Library

```
print(D2)
{'name': 'John Doe', 'DoB': 'March 2nd, 2005', 'grades': ['A+', 'B', 'A-'],
 'year': 2024}
```

```
D2.pop('DoB')
```

```
D2
{'name': 'Tony Clifton',
 'grades': ['A+', 'B', 'A-'],
 'year': 2024,
 'Address': '134 Street, Home'}
```



removing keys



Basic Types

dict

Modules

The math Library

dictionaries are **mutable** too!

```
print(D2)
{'name': 'John Doe', 'DoB': 'March 2nd, 2005', 'grades': ['A+', 'B', 'A-'],
'year': 2024}
```

```
D3 = D2
```

```
D2['year'] = 2025
```

```
D3['year']
2025
```

```
D3 = D2.copy()
```

```
D2['year'] = 2024
```

```
D3['year']
2025
```



Basic Types

set

Modules

The math Library

when to use:

making code more compact (vs control structures)

comparing data entries/ removing duplicates

```
S1 = set(('Mike', 'Karen', 'Simon', 1))
```

```
S2 = set(['Mike', 'Karen', 'Simon', 1])
```

creating a set using
the **constructor** `set`

```
type(S1)
```

```
type(S2)
```

```
set
```

```
set
```

```
S3 = {'Mike', 'Karen', 'Simon', 1}
```

creating a set using
{ }

```
type(S3)
```

```
set
```




Basic Types

set

Modules

The math Library

when to use:

making code more compact (vs control structures)
comparing data entries/ removing duplicates

Note:

sets are **not subscriptable** → `S1[1]` prompts a type error!

duplicates are not permitted

```
S2 = set(['Mike', 'Karen', 'Simon', 1, 1])
```

```
print(S2)
```

```
{1, 'Simon', 'Mike', 'Karen'}
```

sets are **unchangeable**

sets are **mutable**



Basic Types

set

Modules

The math Library

```
S1 = set(('a', 'b', 'c'))
```

```
S2 = set(('a', 'b', 'd'))
```

```
S1.intersection(S2)  
{'a', 'b'}
```



```
S1-S2  
Out[34]: {'c'}
```



```
S2-S1  
Out[35]: {'d'}
```





Basic Types

tuple

Modules

The math Library

when to use:

later: shape of arrays (matrices, data frames)
convenient way to store different objects

```
T1 = tuple([1, 2, 'abc'])
```

```
type(T1)  
tuple
```

creating a tuple using
the **constructor** **tuple**

```
T2 = (1, 2, 'abc')  
type(T2)  
tuple
```

creating a tuple using
()



Basic Types

tuple

Modules

The math Library

```
T1 = tuple([1, 2, 'abc'])
```

```
T2 = (1, 2, 'abc')
```

```
T1[2]  
'abc'
```

indexing & slicing

```
T1[:2]  
(1, 2)
```

```
type(T1[:2])  
tuple
```

```
(t11, t12, t13) = T1
```

retrieving elements

```
print(t11,t12,t13)  
1 2 abc
```



Basic Types

tuple

Modules

The math Library

```
T1 = tuple([1, 2, 'abc'])
```

```
T2 = (1, 2, 'abc')
```

Note:

tuples are **subscriptable** → `T1[1] = 2`

duplicates **are** permitted

tuples are **unchangeable**



Basic Types

Modules

The math Library

summary data collection types

type	constructor	direct construction	mutable	changeable	indexing	slicing	duplicates
list	list	<code>[]</code>	yes	yes	yes	yes	yes
dictionary	dict	<code>{key: value }</code>	yes	yes	no	no	yes
set	set	<code>{}</code>	yes	no	no	no	no
tuple	tuple	<code>()</code>	no	no	yes	yes	yes



The zoo of types



numeric: `int`, `float`, `complex`

`5`, `5.55`, `(5+5j)`

strings: `str`

`'this is a string'`, `"this is a string"`

iteratable

sequence: `list`, `tuple`, `range`

`my_tuple = (3, 'a', [2,3,4,5])`
`range(10)`

`my_list = [1, 2, 'a']`



mapping

mapping: `dict`

`my_dict = {1: 'a', 2: 'b'}`

mapping: `set`

`my_set = {1, 2, 'a'}`



boolean:

`True` `False`

none type:

`None`

callable: functions, methods, classes

`def`, `class`, `map`, `lambda`

modules:

`from my_module import my_method as my_alias`



Basic Types
Modules
The math Library

```
from my_module import my_method as my_alias
```

1) reading files (.xlsx, .xls, .csv, .txt, ...)

pandas (standard), dask, polars

2) plotting

matplotlib, seaborn

3) numerical methods

math, numpy, scipy

4) machine learning

scikitlearn

5) ANN/AI/DeepLearning


TensorFlow

 Keras

 PyTorch



Basic Types

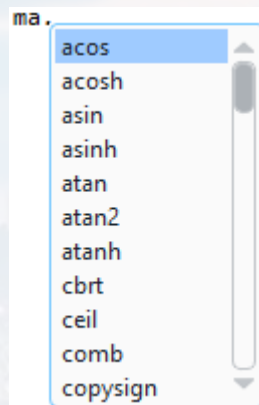
Modules

The math Library

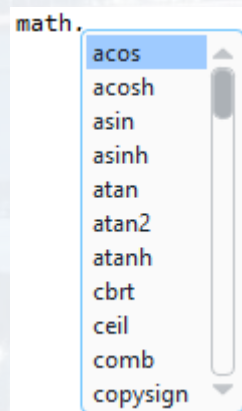
```
from my_module import my_method as my_alias
```

alias

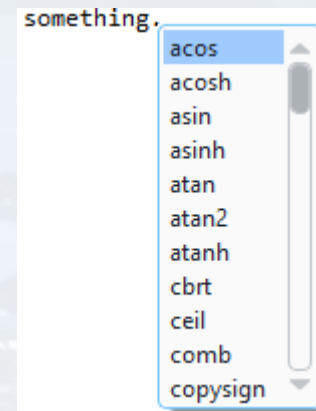
```
import math as ma
```



```
import math as math
```



```
import math as something
```





Basic Types
Modules
The math Library

importing
specific tools

```
from my_module import my_method as my_alias
```

```
from math import cos as cosine
```

```
cosine(3.14159)  
-0.999999999999964793
```

```
sin(3.14159)
```

```
Traceback (most recent call last):
```

```
Cell In[11], line 1  
    sin(3.14159)
```

```
NameError: name 'sin' is not defined
```

the method sin has not been imported yet



Basic Types

Modules

The math Library

importing
specific tools

```
from my_module import my_method as my_alias
```

```
from math import cos, sin
```

```
cos(3.14159)  
-0.99999999999964793
```

```
sin(3.14159)  
2.65358979335273e-06
```



Basic Types
Modules
The math Library

```
from my_module import my_method as my_alias
```

importing
specific tools

```
from math import cos, sin
```

```
cos(3.14159)  
-0.99999999999964793
```

```
sin(3.14159)  
2.65358979335273e-06
```

importing all
tools at once

```
from math import *
```

```
cos(3.14159)  
-0.99999999999964793
```

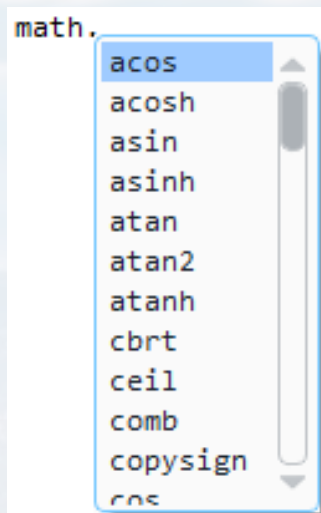
```
sin(3.14159)  
2.65358979335273e-06
```

```
tan(3.14159)  
-2.6535897933620727e-06
```




Basic Types
Modules
The math Library

```
import math as math
```



```
math.acos(0) * 2  
3.141592653589793
```

importing a module as alias
→ calling a method via
my_alias.method, i.e.

```
math.acos  
math.ceil
```

or

```
from math import *
```

```
acos(0) * 2  
3.141592653589793
```

importing all methods (*)
from a module

```
acos  
ceil
```



Basic Types
Modules
The math Library

```
import math as math
```

math contains a vast set of mathematical operations

```
dir(math)
```

```
'acos', 'acosh', 'asin', 'asinh', 'atan', 'atan2', 'atanh', 'cbrt', 'ceil', 'comb', 'copysign', 'cos', 'cosh', 'degrees', 'dist', 'e', 'erf', 'erfc', 'exp', 'exp2', 'expm1', 'fabs', 'factorial', 'floor', 'fmod', 'frexp', 'fsum', 'gamma', 'gcd', 'hypot', 'inf', 'isclose', 'isfinite', 'isinf', 'isnan', 'isqrt', 'lcm', 'ldexp', 'lgamma', 'log', 'log10', 'log1p', 'log2', 'modf', 'nan', 'nextafter', 'perm', 'pi', 'pow', 'prod', 'radians', 'remainder', 'sin', 'sinh', 'sqrt', 'tan', 'tanh', 'tau', 'trunc', 'ulp']
```

each method includes a documentation

```
math.log(  
    log(x, [base=math.e])  
    Return the logarithm of x to the given base.  
    If the base not specified, returns the natural logarithm  
    (base e) of x.
```



Basic Types
Modules
The math Library

```
import math as math
```

Homework assignment!



Calculate the values for the following equations using math

$$\log_4(32)$$

$$\cos(60^\circ)$$

$$\sqrt{2 + 5i}$$

$$e^{i\pi}$$

$$\frac{e^5}{6!}$$

$$\binom{10}{5}$$



Thank you for your attention!